Report Generated by Test Manager

Title: Test Reports of SoC
Author: Victoria
Date: 10-Apr-2026 11:20:48

Test Environment

Platform: PCWIN64
MATLAB: (R2025b)

Summary

Name	Outcome	Duration (Seconds)
Results: 2026-Apr-10 09:50:20		3.36
worst_case_test		2.81
Results: 2026-Apr-10 09:56:41		1.982
battery_voltage_test		1.717
Results: 2026-Apr-10 09:58:09		1.621
cell_temperature_test		1.33
Results: 2026-Apr-10 09:59:21		1.724
low_battery_test		1.532
Results: 2026-Apr-10 11:20:21		5.793
soc_tolerance_band_test		3.993

Results: 2026-Apr-10 09:50:20

Result Type:	Result Set
Parent:	None
Start Time:	10-Apr-2026 09:50:21
End Time:	10-Apr-2026 09:50:24
Outcome:	Total: 1, Passed: 1

[Back to Report Summary]

worst_case_test

Test Result Information

Result Type:	Test Case Result
Parent:	Results: 2026-Apr-10 09:50:20
Start Time:	10-Apr-2026 09:50:21
End Time:	10-Apr-2026 09:50:24
Outcome:	Passed

Test Case Information

Name:	worst_case_test
Type:	Simulation Test

Logical and Temporal Assessments

Name	Assessment
Assessment1	At any point in time, if (((cell_temperature > 90) & (battery_voltage <= 30)) & (soc < 20)) becomes true then, with a delay of at most 1 seconds, ((status_flag == true) & (low_battery_warnflag == true)) must be true

Simulation

System Under Test Information

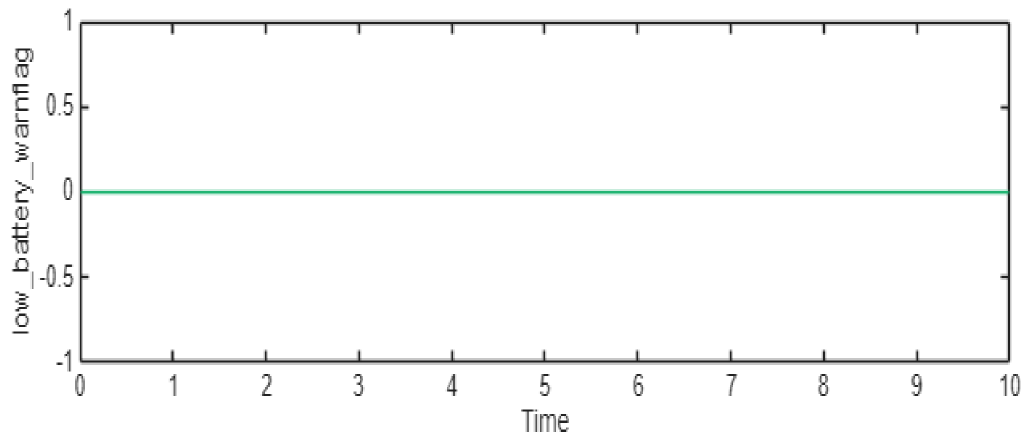
Model:	SOC_Regler
Release:	Current
Simulation Mode:	normal
Override SIL or PIL Mode:	0
Configuration Set:	Configuration

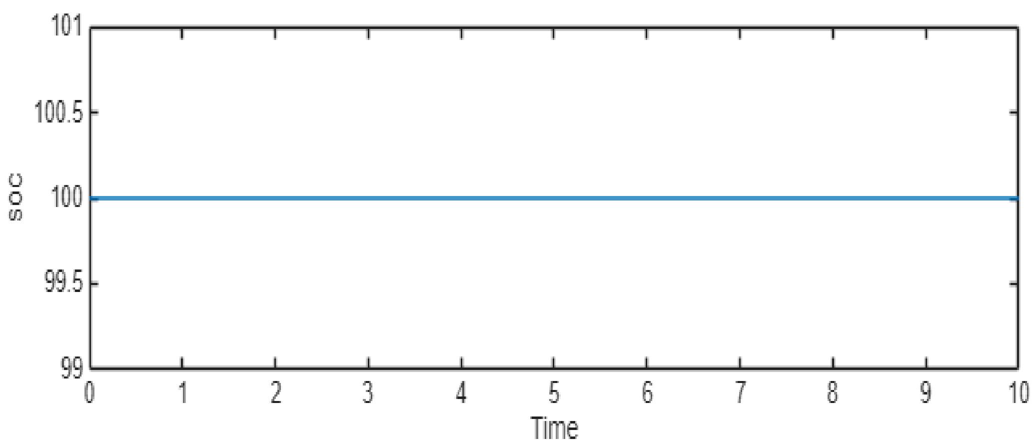
External Input Name:	worst_case_input.mat
External Input File:	C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\04_Verification\worst_case_input.mat
Start Time:	0
Stop Time:	10
Checksum:	3233302630 925315875 422518952 2889595808
Simulink Version:	25.2
Model Version:	1.4
Model Author:	320236370
Date:	Wed Apr 08 19:47:39 2026
User ID:	320236370
Model Path:	C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\SOC_Regler.slx
Machine Name:	YY309528
Solver Name:	ode45
Solver Type:	Variable-Step
Max Step Size:	0.20000000000000001
Simulation Start Time:	2026-04-10 09:50:21
Simulation Stop Time:	2026-04-10 09:50:22
Platform:	PCWIN64

Simulation Output

Name	Data Type	Units	Sample Time	Interp	Sync	Link to Plot
low_battery_warnflag	boolean		Continuous	zoh	union	Link
soc	double		Continuous	linear	union	Link
low_battery_warnflag	boolean		Continuous	zoh	union	Link
soc	double		Continuous	linear	union	Link
status_flag	boolean		Continuous	zoh	union	Link
battery_voltage	double		Continuous	linear	union	Link
cell_temperature	double		Continuous	linear	union	Link
status_flag	boolean		Continuous	zoh	union	Link

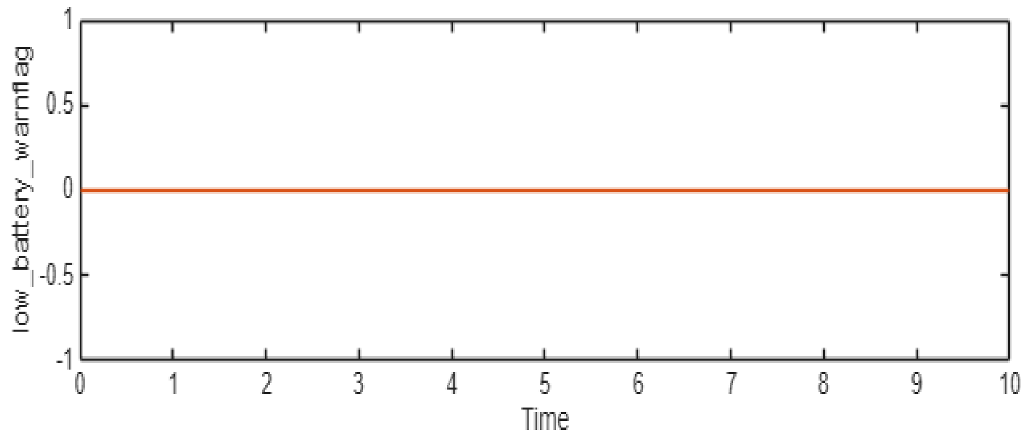
Name	Data Type	Units	Sample Time	Interp	Sync
low_battery_warnflag	boolean		Continuous	zoh	union
soc	double		Continuous	linear	union

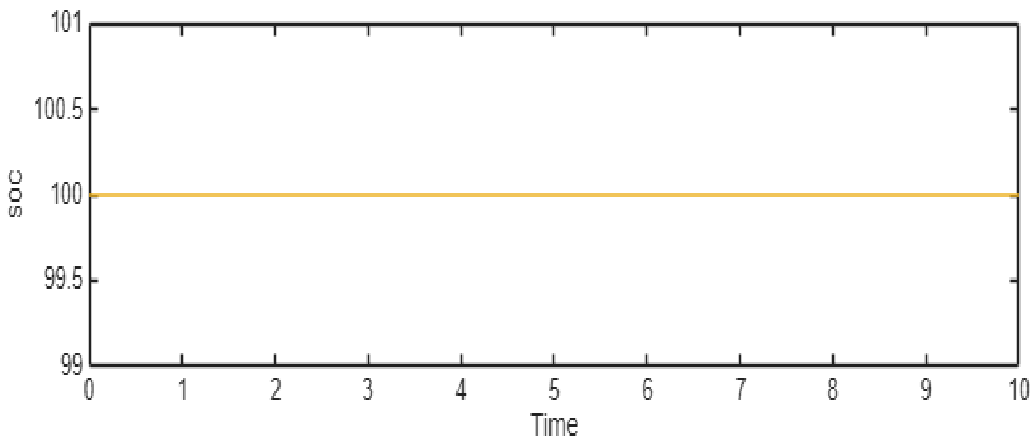




[Back to Report Summary](#)[Back to Signal Summary](#)

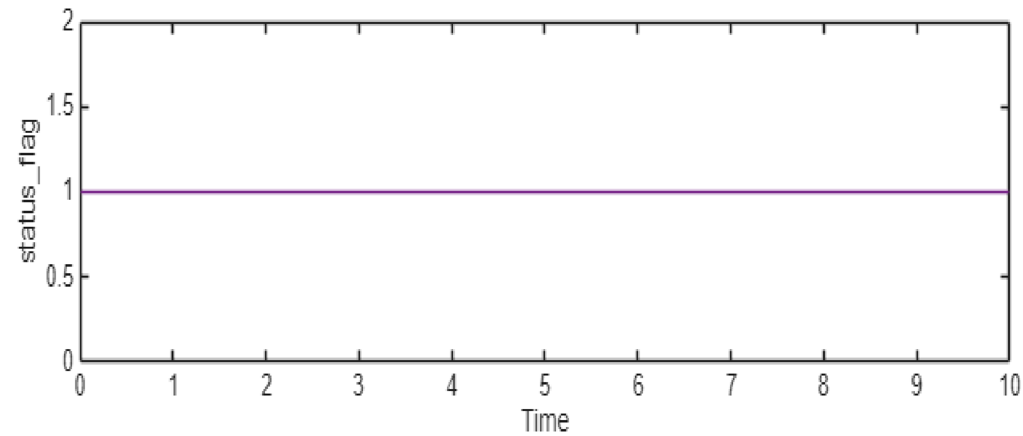
Name	Data Type	Units	Sample Time	Interp	Sync
low_battery_warnflag	boolean		Continuous	zoh	union
soc	double		Continuous	linear	union

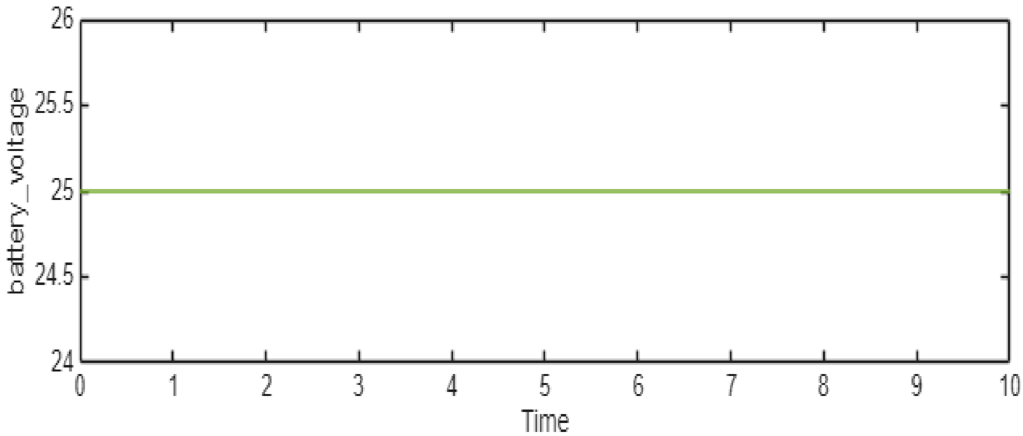




Back to Report SummaryBack to Signal Summary

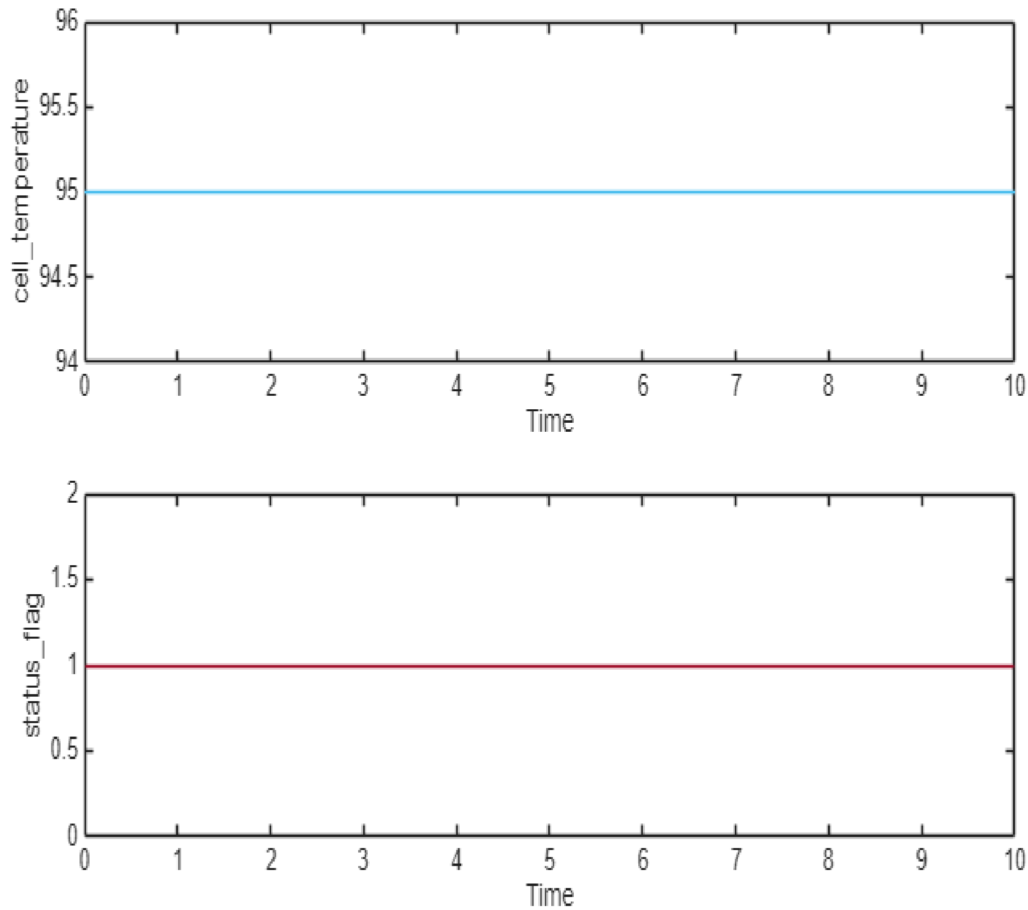
Name	Data Type	Units	Sample Time	Interp	Sync
status_flag	boolean		Continuous	zoh	union
battery_voltage	double		Continuous	linear	union





Back to Report SummaryBack to Signal Summary

Name	Data Type	Units	Sample Time	Interp	Sync
cell_temperature	double		Continuous	linear	union
status_flag	boolean		Continuous	zoh	union



Back to Report SummaryBack to Signal Summary

Simulation Logs:
Inputs may not be compatible for simulation. Test results might not be accurate. Click here for more information on external input mapping.

Back to Report Summary

Results: 2026-Apr-10 09:56:41

Result Type:	Result Set
Parent:	None
Start Time:	10-Apr-2026 09:56:41
End Time:	10-Apr-2026 09:56:43
Outcome:	Total: 1, Passed: 1

Back to Report Summary

battery_voltage_test

Test Result Information

Result Type:	Test Case Result
Parent:	Results: 2026-Apr-10 09:56:41
Start Time:	10-Apr-2026 09:56:41
End Time:	10-Apr-2026 09:56:43
Outcome:	Passed

Test Case Information

Name:	battery_voltage_test
Type:	Simulation Test

Logical and Temporal Assessments

Name	Assessment
Assessment1	At any point in time, if (battery_voltage <= 30) becomes true then, with a delay of at most 1 seconds, (status_flag == true) must be true

Simulation

System Under Test Information

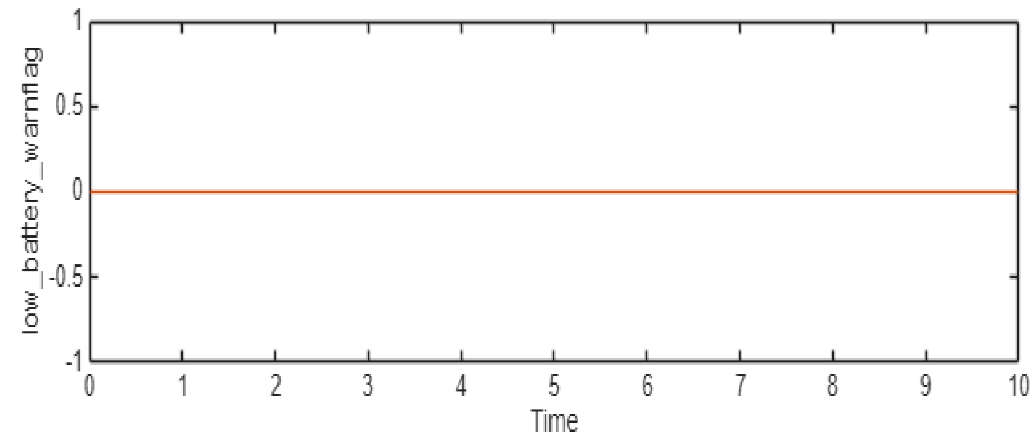
Model:	SOC_Regler
Release:	Current
Simulation Mode:	normal

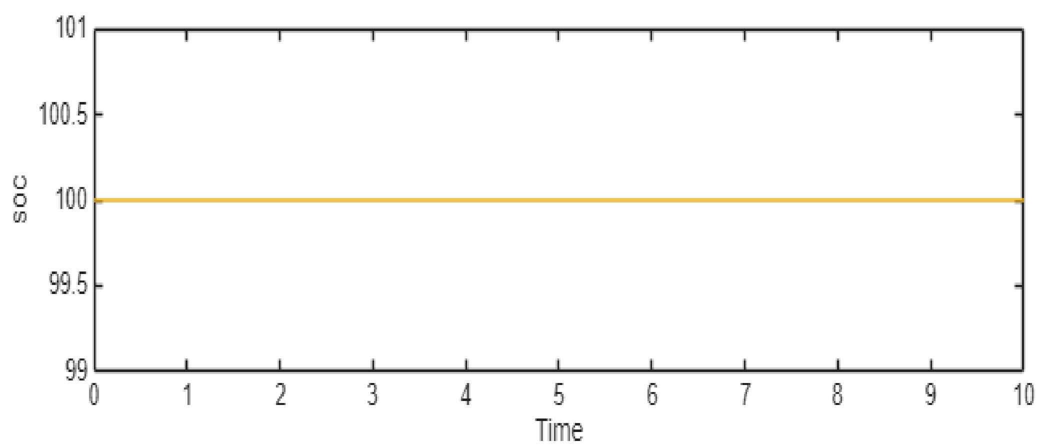
Override SIL or PIL Mode:	0
Configuration Set:	Configuration
External Input Name:	battery_voltage_input.mat (1)
External Input File:	C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\04_Verification\input_data\battery_voltage_input.mat
Start Time:	0
Stop Time:	10
Checksum:	4132136549 3765386925 1652960231 359931920
Simulink Version:	25.2
Model Version:	1.4
Model Author:	320236370
Date:	Wed Apr 08 19:47:39 2026
User ID:	320236370
Model Path:	C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\SOC_Regler.slx
Machine Name:	YY309528
Solver Name:	ode45
Solver Type:	Variable-Step
Max Step Size:	0.20000000000000001
Simulation Start Time:	2026-04-10 09:56:41
Simulation Stop Time:	2026-04-10 09:56:42
Platform:	PCWIN64

Simulation Output

Name	Data Type	Units	Sample Time	Interp	Sync	Link to Plot
low_battery_warnflag	boolean		Continuous	zoh	union	Link
soc	double		Continuous	linear	union	Link
status_flag	boolean		Continuous	zoh	union	Link
battery_voltage	double		Continuous	linear	union	Link
status_flag	boolean		Continuous	zoh	union	Link

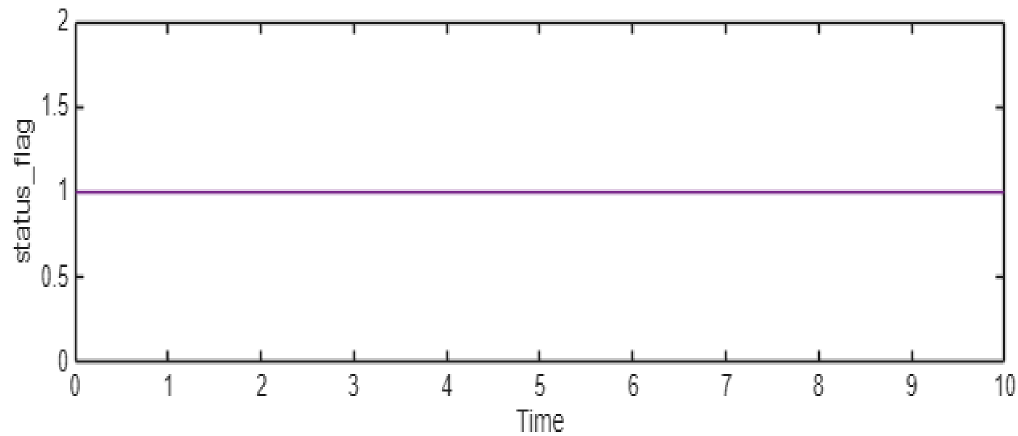
Name	Data Type	Units	Sample Time	Interp	Sync
low_battery_warnflag	boolean		Continuous	zoh	union
soc	double		Continuous	linear	union

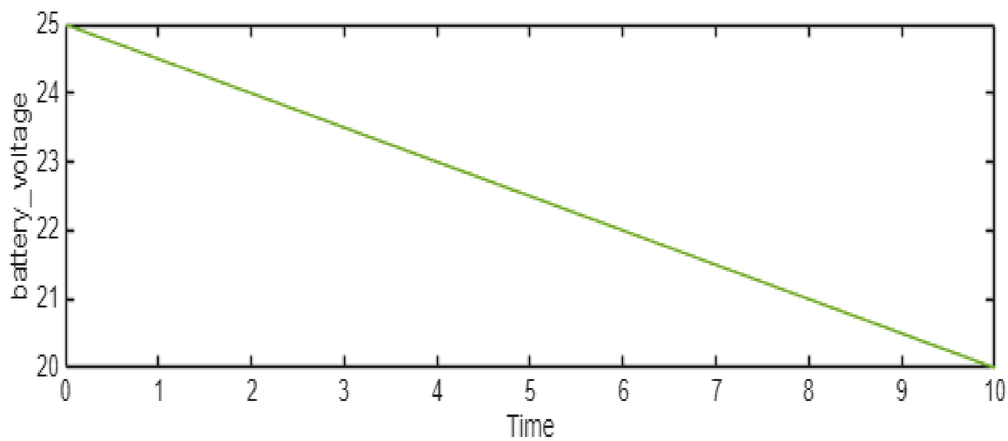




[Back to Report Summary](#)[Back to Signal Summary](#)

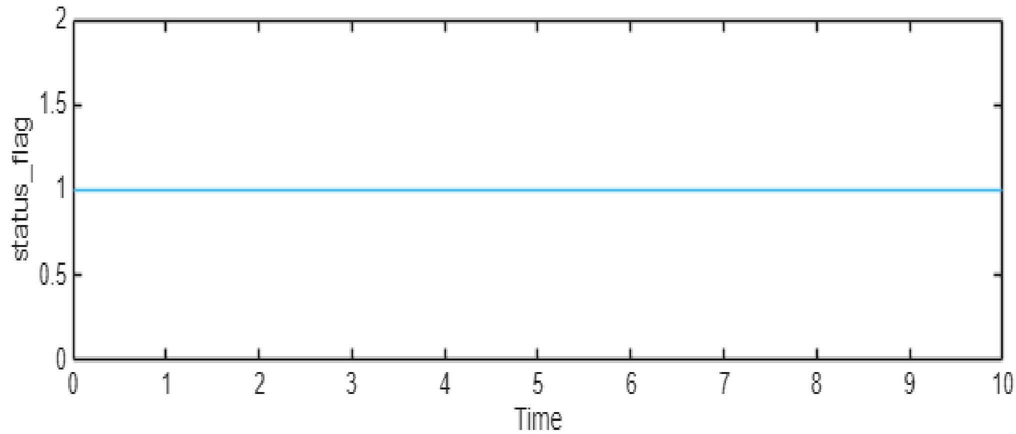
Name	Data Type	Units	Sample Time	Interp	Sync
status_flag	boolean		Continuous	zoh	union
battery_voltage	double		Continuous	linear	union





Back to Report SummaryBack to Signal Summary

Name	Data Type	Units	Sample Time	Interp	Sync
status_flag	boolean		Continuous	zoh	union



Back to Report SummaryBack to Signal Summary

Simulation Logs:
Inputs may not be compatible for simulation. Test results might not be accurate. Click here for more information on external input mapping.

Back to Report Summary

Results: 2026-Apr-10 09:58:09

Result Type:	Result Set
Parent:	None
Start Time:	10-Apr-2026 09:58:09
End Time:	10-Apr-2026 09:58:10
Outcome:	Total: 1, Passed: 1

Back to Report Summary

cell_temperature_test

Test Result Information

Result Type:	Test Case Result
Parent:	Results: 2026-Apr-10 09:58:09
Start Time:	10-Apr-2026 09:58:09
End Time:	10-Apr-2026 09:58:10
Outcome:	Passed

Test Case Information

Name:	cell_temperature_test
Type:	Baseline Test

Logical and Temporal Assessments

Name	Assessment
Assessment1	At any point in time, if (cell_temperature > 90) becomes true then, with a delay of at most 1 seconds, (status_flag == true) must be true

Simulation

System Under Test Information

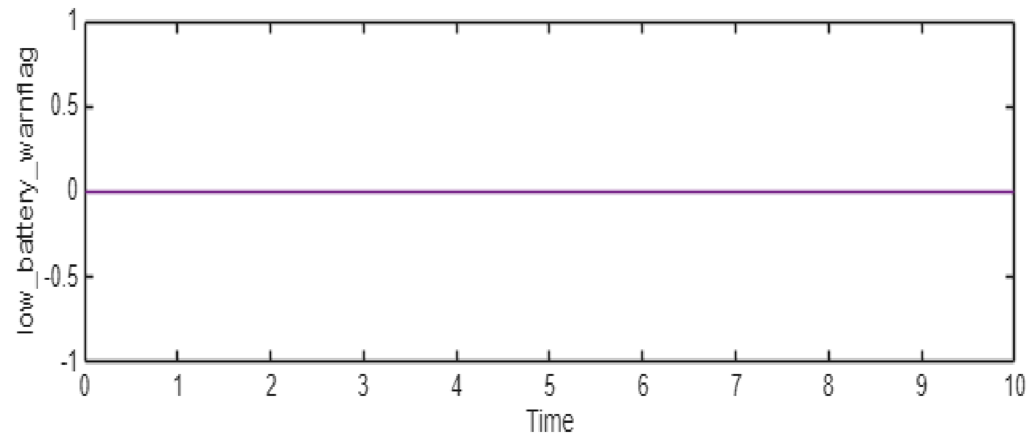
Model:	SOC_Regler
Release:	Current
Simulation Mode:	normal

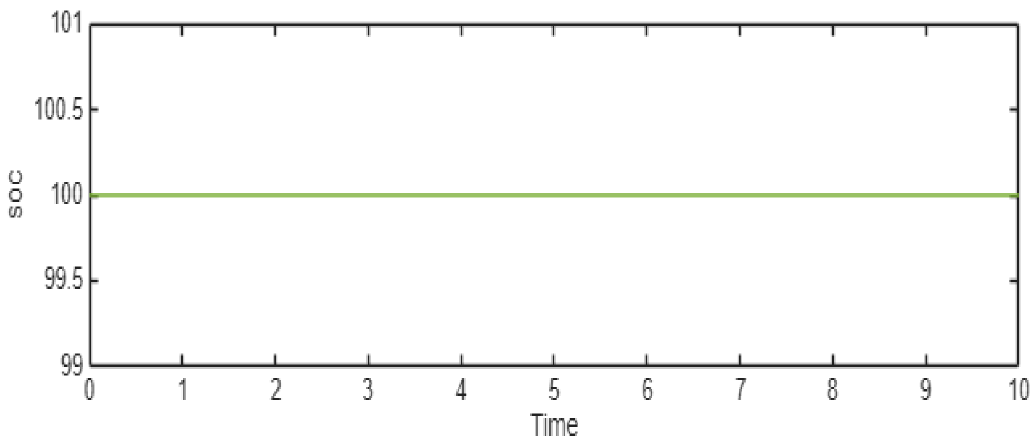
Override SIL or PIL Mode:	0
Configuration Set:	Configuration
External Input Name:	cell_temperature_input.mat (1)
External Input File:	C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\04_Verification\input_data\cell_temperature_input.mat
Start Time:	0
Stop Time:	10
Checksum:	1396786863 1825109906 3322312545 4255407849
Simulink Version:	25.2
Model Version:	1.4
Model Author:	320236370
Date:	Wed Apr 08 19:47:39 2026
User ID:	320236370
Model Path:	C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\SOC_Regler.slx
Machine Name:	YY309528
Solver Name:	ode45
Solver Type:	Variable-Step
Max Step Size:	0.20000000000000001
Simulation Start Time:	2026-04-10 09:58:09
Simulation Stop Time:	2026-04-10 09:58:09
Platform:	PCWIN64

Simulation Output

Name	Data Type	Units	Sample Time	Interp	Sync	Link to Plot
low_battery_warnflag	boolean		Continuous	zoh	union	Link
soc	double		Continuous	linear	union	Link
status_flag	boolean		Continuous	zoh	union	Link
cell_temperature	double		Continuous	linear	union	Link
status_flag	boolean		Continuous	zoh	union	Link

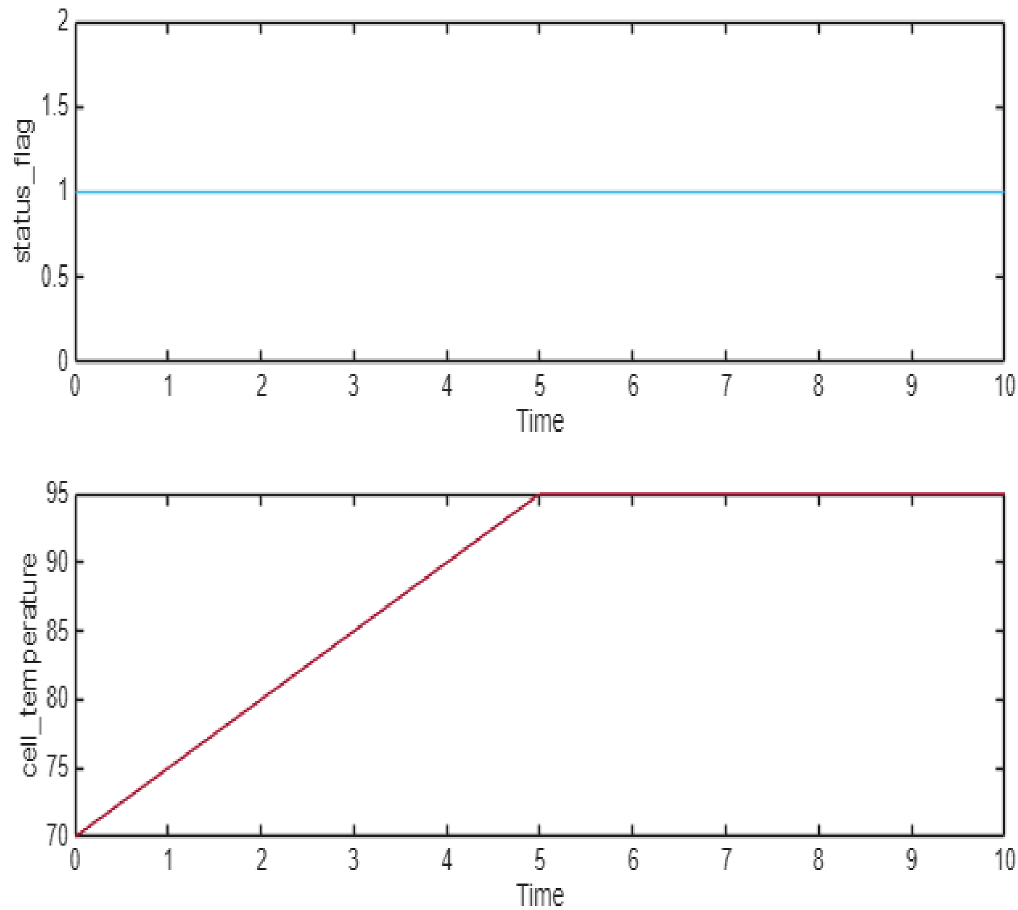
Name	Data Type	Units	Sample Time	Interp	Sync
low_battery_warnflag	boolean		Continuous	zoh	union
soc	double		Continuous	linear	union





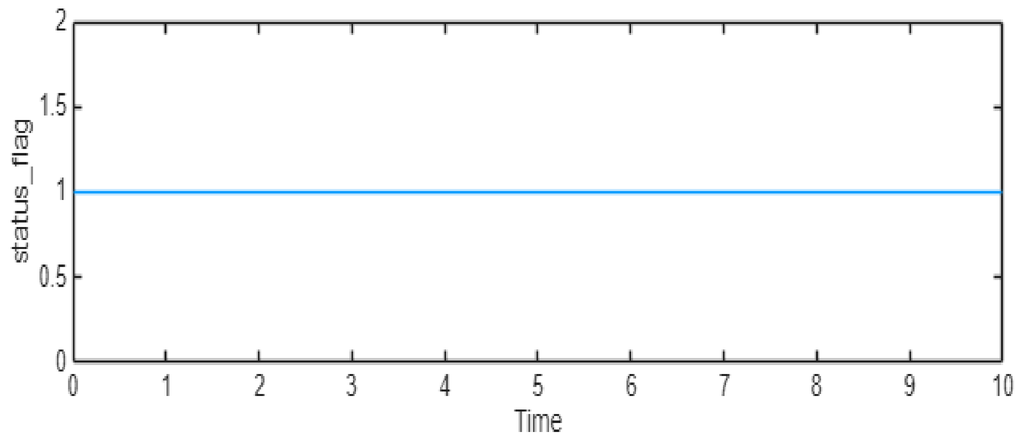
[Back to Report Summary](#)[Back to Signal Summary](#)

Name	Data Type	Units	Sample Time	Interp	Sync
status_flag	boolean		Continuous	zoh	union
cell_temperature	double		Continuous	linear	union



[Back to Report Summary](#)[Back to Signal Summary](#)

Name	Data Type	Units	Sample Time	Interp	Sync
status_flag	boolean		Continuous	zoh	union



Back to Report SummaryBack to Signal Summary

Simulation Logs:
Inputs may not be compatible for simulation. Test results might not be accurate. Click here for more information on external input mapping.

Back to Report Summary

Test Logs:
No baseline criteria evaluation performed as no baseline data is available for this test.

Back to Report Summary

Results: 2026-Apr-10 09:59:21

Result Type:	Result Set
Parent:	None
Start Time:	10-Apr-2026 09:59:21
End Time:	10-Apr-2026 09:59:22
Outcome:	Total: 1, Passed: 1

Back to Report Summary

low_battery_test

Test Result Information

Result Type:	Test Case Result
Parent:	Results: 2026-Apr-10 09:59:21
Start Time:	10-Apr-2026 09:59:21
End Time:	10-Apr-2026 09:59:22
Outcome:	Passed

Test Case Information

Name:	low_battery_test
Type:	Simulation Test

Logical and Temporal Assessments

Name	Assessment
Assessment1	At any point in time, if (state_of_current < 20) becomes true then, with a delay of at most 1 seconds, (low_battery_warnflag > 0.5) must be true

Simulation

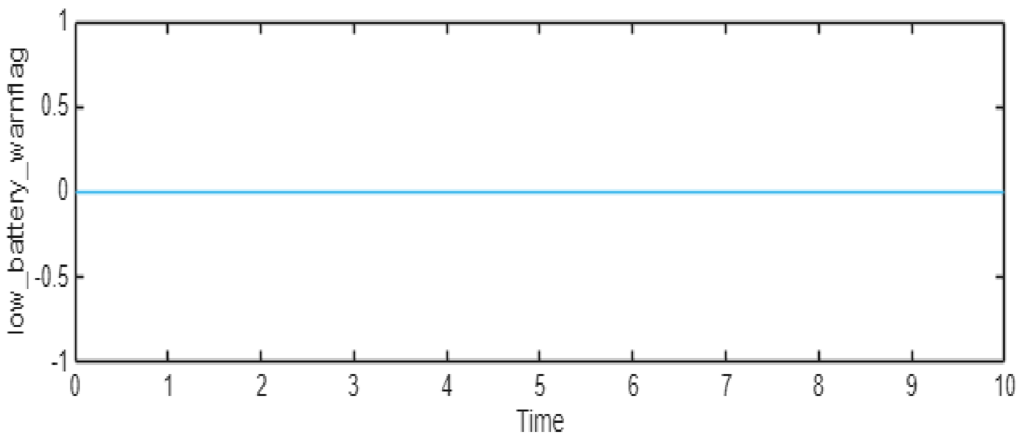
System Under Test Information

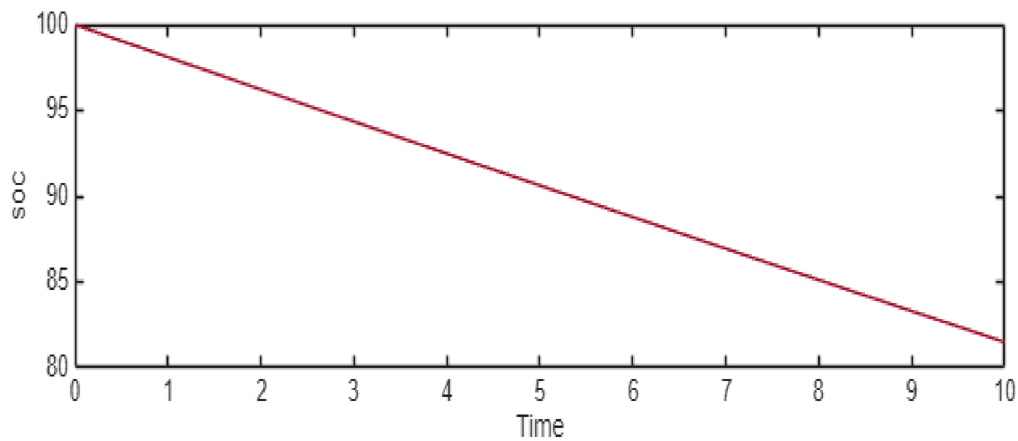
Model:	SOC_Regler
Release:	Current
Simulation Mode:	normal
Override SIL or PIL Mode:	0
Configuration Set:	Configuration
External Input Name:	battery_current_input.mat
External Input File:	C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\04_Verification\input_data\battery_current_input.mat
Start Time:	0
Stop Time:	10
Checksum:	1580065241 1310189143 2029611893 136738514
Simulink Version:	25.2
Model Version:	1.4
Model Author:	320236370
Date:	Wed Apr 08 19:47:39 2026
User ID:	320236370
Model Path:	C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\SOC_Regler.slx
Machine Name:	YY309528
Solver Name:	ode45
Solver Type:	Variable-Step
Max Step Size:	0.20000000000000001
Simulation Start Time:	2026-04-10 09:59:21
Simulation Stop Time:	2026-04-10 09:59:22
Platform:	PCWIN64

Simulation Output

Name	Data Type	Units	Sample Time	Interp	Sync	Link to Plot
low_battery_warnflag	boolean		Continuous	zoh	union	Link
soc	double		Continuous	linear	union	Link
soc	double		Continuous	linear	union	Link
status_flag	boolean		Continuous	zoh	union	Link

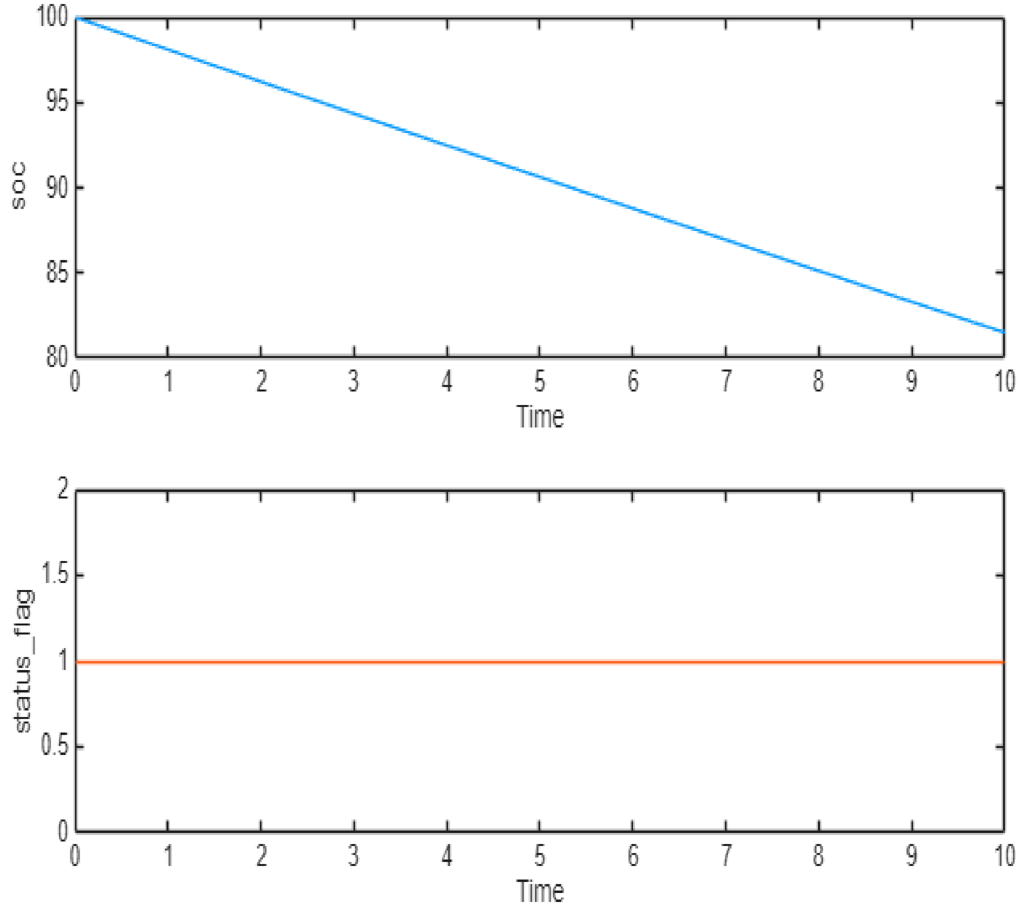
Name	Data Type	Units	Sample Time	Interp	Sync
low_battery_warnflag	boolean		Continuous	zoh	union
soc	double		Continuous	linear	union





Back to Report SummaryBack to Signal Summary

Name	Data Type	Units	Sample Time	Interp	Sync
soc	double		Continuous	linear	union
status_flag	boolean		Continuous	zoh	union



Back to Report SummaryBack to Signal Summary

Simulation Logs:
Inputs may not be compatible for simulation. Test results might not be accurate. Click here for more information on external input mapping.

Back to Report Summary

Results: 2026-Apr-10 11:20:21

Result Type:	Result Set
Parent:	None
Start Time:	10-Apr-2026 11:20:22
End Time:	10-Apr-2026 11:20:28
Outcome:	Total: 1, Passed: 1

[Back to Report Summary]

soc_tolerance_band_test

Test Result Information

Result Type:	Test Case Result
Parent:	Results: 2026-Apr-10 11:20:21
Start Time:	10-Apr-2026 11:20:22
End Time:	10-Apr-2026 11:20:26
Outcome:	Passed

Test Case Information

Name:	soc_tolerance_band_test
Type:	Simulation Test

Test Case Requirements

Description:	BAT_CTRL_1 Toleranz der Berechnung
Document:	Regler_Requirements.slreqx

Logical and Temporal Assessments

Name	Assessment
Assessment1	At any point in time, if (battery_current == 0) becomes true then, with a delay of at most 1 seconds, (soc == 0) must be true

Simulation

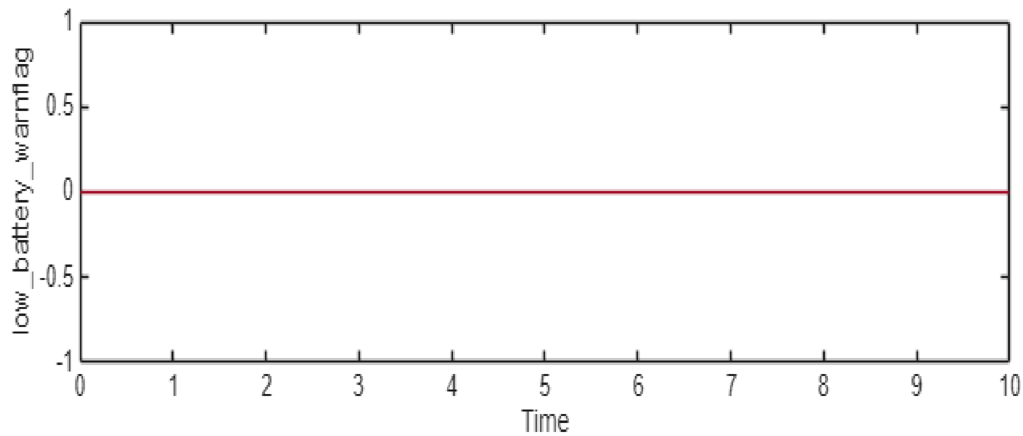
System Under Test Information

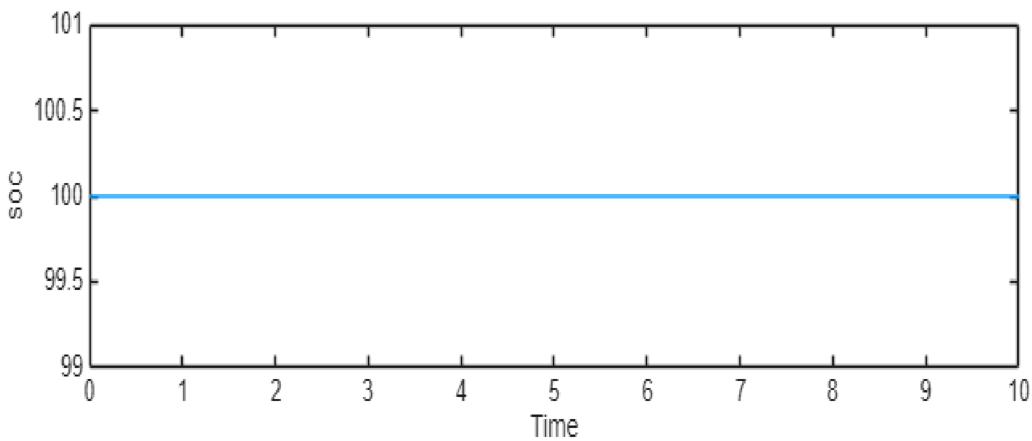
Model:	SOC_Regler
Release:	Current
Simulation Mode:	normal
Override SIL or PIL Mode:	0
Configuration Set:	Configuration
External Input Name:	battery_current_0_input.mat
External Input File:	C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\04_Verification\battery_current_0_input.mat
Start Time:	0
Stop Time:	10
Checksum:	3846692330 2512201270 814111440 1202636617
Simulink Version:	25.2
Model Version:	1.4
Model Author:	320236370
Date:	Wed Apr 08 19:47:39 2026
User ID:	320236370
Model Path:	C:\Users\320236370\Downloads\Escooter-MBSE-System-Model-main\Escooter-MBSE-System-Model-main\SOC_Regler.slx
Machine Name:	YY309528
Solver Name:	ode45
Solver Type:	Variable-Step
Max Step Size:	0.20000000000000001
Simulation Start Time:	2026-04-10 11:20:22
Simulation Stop Time:	2026-04-10 11:20:25
Platform:	PCWIN64

Simulation Output

Name	Data Type	Units	Sample Time	Interp	Sync	Link to Plot
low_battery_warnflag	boolean		Continuous	zoh	union	Link
soc	double		Continuous	linear	union	Link
battery_current	double		Continuous	linear	union	Link
soc	double		Continuous	linear	union	Link
status_flag	boolean		Continuous	zoh	union	Link

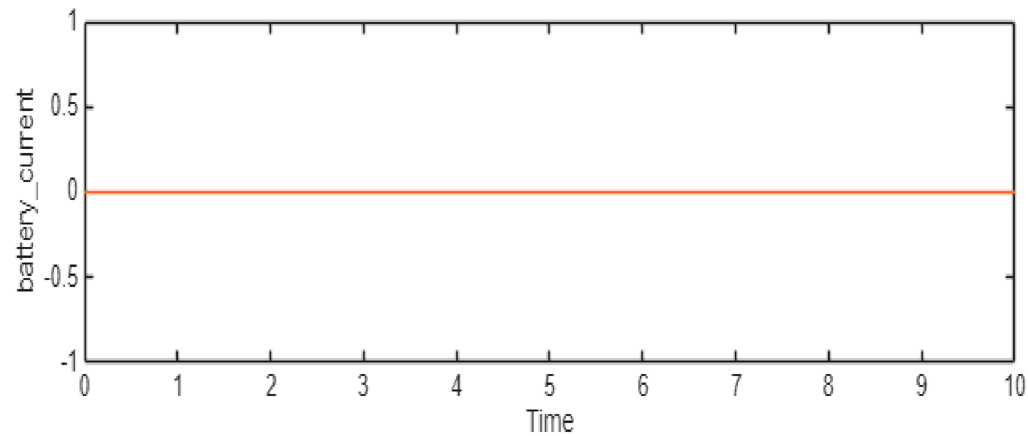
Name	Data Type	Units	Sample Time	Interp	Sync
low_battery_warnflag	boolean		Continuous	zoh	union
soc	double		Continuous	linear	union

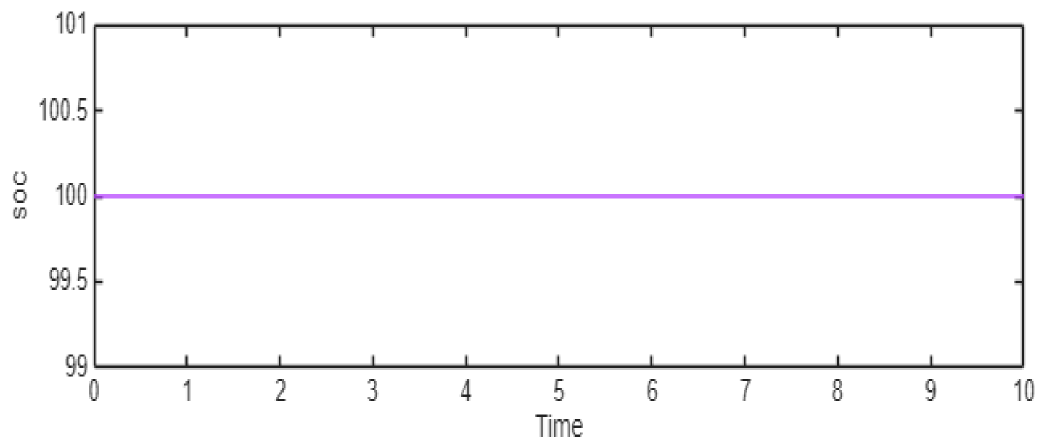




Back to Report SummaryBack to Signal Summary

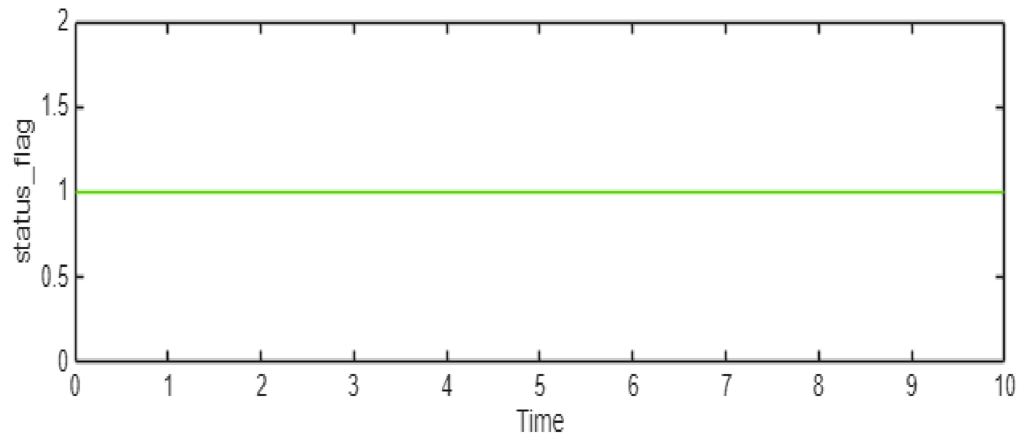
Name	Data Type	Units	Sample Time	Interp	Sync
battery_current	double		Continuous	linear	union
soc	double		Continuous	linear	union





Back to Report SummaryBack to Signal Summary

Name	Data Type	Units	Sample Time	Interp	Sync
status_flag	boolean		Continuous	zoh	union



Back to Report SummaryBack to Signal Summary

Simulation Logs:
Inputs may not be compatible for simulation. Test results might not be accurate. Click here for more information on external input mapping.

Back to Report Summary